This project is an attempt to visualize the Japanese National Diet Database data related to Taiwan, using the free API provided by the National Diet Library of Japan (https://teikokugikai-i.ndl.go.jp/)

The dataset consists of extracted speech excerpts and full text records spanning 1894 to 1945 with a focus on Taiwan-related discussions during the Japanese colonial period. The main goal of this project is to explore how Taiwan is represented in parliamentary discourse over time and how these representations change in response to major political and historical events. The data (all the speeches that contain the word "Taiwan" from 1890-1947) was preprocessed from scraped and structured sources into CSV files to be further processed and presented.

A key part of the project is natural language processing using SudachiPy for Japanese tokenization. The text was normalized by converting katakana to hiragana and then tokenized using Sudachi in mode C. The analysis focuses on extracting nouns as the main semantic units. This design choice was made because nouns tend to carry topical meaning in historical political texts. However, one important limitation is that the noun filtering process is not perfect. Many functional or weakly meaningful nouns are still present in the output despite the use of a manually designed stopword list. This means that the word cloud results still include noise and could be improved in future work by using more advanced filtering methods. One choice is to allow Kanji only. However, this may also lead to a loss of useful information. Despite this limitation, noun-based extraction was chosen because it provides a simple, interpretable, and computationally efficient way to explore large historical text data within the time constraints of the project.

The main visualization in this project is a word cloud by year. The purpose of this visualization is not to produce precise linguistic conclusions but to provide an intuitive overview of dominant terms and themes in each time period.

In addition to the word cloud, the project also includes document distribution charts showing the number of documents per year and per decade. These visualizations are important because they help contextualize the dataset size over time and prevent overinterpretation of individual years or isolated spikes in word frequency. Without this context, it would be easy to draw misleading conclusions from years with unusually high or low document counts. Furthermore, the presence or absence of “Taiwan” itself is useful information. Therefore, these charts serve as a normalization and interpretation aid for the rest of the dashboard.

A timeline panel is also included to help interpret word cloud patterns. The panel contains historical events and administrative changes, such as shifts in governance policy and major incidents in Taiwan during the colonial period. This can be expanded further in the future. Its present shape is to demonstrate the usefulness of such an aid.

During development, I also attempted to incorporate a historical map visualization to provide geographic context for the events and text references. The idea was to make the dashboard more spatially aware by connecting textual discourse with the geographic representation of Taiwan and surrounding regions. However, the map component remains relatively basic and does not yet fully represent dynamic historical spatial changes. To properly implement this feature, a transition to a more flexible framework, such as Dash, may be necessary in the future since it would allow more complex interactive map controls and layered historical visualizations. In the end, I did not include the display of the historical map. A representation of current-day Taiwan is kept.

This project went through several iterations, including web scraping experiments, multiple versions of preprocessing pipelines, and exploration of more complex visualization methods such as topic modeling and clustering. During the design phase, I also used ChatGPT to help interpret the lecture-provided example code, evaluate the feasibility of different visualization approaches, and structure parts of the Streamlit application. Some parts of the README and documentation were also generated with assistance from AI tools and then manually edited. In the main .py file, I borrowed code from the course example and debugged it using GPT. The most important contribution of GPT is the strategy it suggested for scraping the source. I attempted to scrap everything at once, but kept failing after reaching a limit. GPT suggested scraping by intervals. This proves to be the best strategy. In addition, I tried other Japanese NLPs from a blog post, which worked poorly on pre-war Japanese. and GPT suggested SudachiPy to first transform katakana into hiragana, and then parse. This improved the quality significantly.

Future general directions (from easier to more difficult) may include:

1. Visualization of statistical data related to Indigenous populations and colonial industries between 1895 and 1945 (based on public datasets such as the Taiwan Legal Empirical Database for the Japanese Colonial Period http://tcsd.lib.ntu.edu.tw/ and Statistical Abstract of Taiwan Province https://twstudy.iis.sinica.edu.tw/TwStatistic50/index.htm)
2. Improving on the filtering and historical information of the current project. For example, allow for a different word of focus to be added by the user, which will change the word cloud at the indicated year
3. Creation of an interactive map based on Wulai maps and travel texts from the Japanese colonial period (using materials collected by the National Taiwan Library and the Nichinichi Shinpo database; texts are extracted via OCR and proofread, after which relationship networks are constructed and overlaid onto Google Maps and multilayer historical maps) (Currently working on aligning 1930s Wulai maps with maps from the 1940s. Because map projections at the time were relatively imprecise, this is still being discussed with the advisor.)